# Shuai Zhang

Assistant Professor
Department of Data Science
Ying Wu College of Computing
GITC 2117
Phone: 973–596–2743
Email: sz457@njit.edu


## Education & Professional Experience

- **New Jersey Institute of Technology (NJIT)**, NJ, US — Aug. 2023 – Present

  Assistant Professor, Department of Data Science

- **Rensselaer Polytechnic Institute (RPI)**, NY, US — Jan. 2022 – Aug. 2023

  Postdoctoral Researcher

- **Rensselaer Polytechnic Institute (RPI)**, NY, US — Aug. 2016 – Dec. 2021

  Ph.D., Electrical, Computer, and Systems Engineering

- **University of Science and Technology of China (USTC)**, China — Sep. 2012 – Jul. 2016

  B.Eng., Electrical Engineering

## Research Interests

My research focuses on the learning theory of emerging artificial intelligence technologies and the structural properties of modern models, with an emphasis on understanding their implications for training dynamics and generalization. I aim to develop principled theoretical frameworks that explain how architectural structure, optimization geometry, and data properties interact to shape the performance, efficiency, and reliability of AI systems.

## Publications

**Note: Boldface** indicates **Shuai Zhang**. An asterisk (*) denotes an NJIT student co-author.

Full publication available at: `https://scholar.google.com/citations?user=RvDk-iwAAAAJ&hl`

### A. Authored Books

- Songyang Zhang (Ed.), **Shuai Zhang** (Ed.), Chuan Huang (Ed.). *Generative Learning for Wireless Communications: Fundamentals and Applications.* Elsevier Science Publishing Co. Inc., expected July 1, 2026, 325 pp. ISBN: 978-0443414978.
  Available at: `https://www.amazon.com/dp/B0DQR1YKML`

## B. Refereed Journal Articles

### B.1 Accepted

[J1] Yating Zhou, **Shuai Zhang**, Meng Wang. "Mid-Term Load Forecasting with Minimal Data: An In-Context-Learning-Aware Approach Using Large Language Models." *IEEE Transactions on Power Systems*, **Accepted**, 2026. *(Peer-reviewed)*

[J2] Jingzehua Xu, Guanwen Xie, Jiwei Tang, Yimian Ding, Weiyi Liu, Junhao Huang, **Shuai Zhang**, Yi Li. "Never Too Cocky to Cooperate: A Fisher-Information-Matrix- and RL-Based USV–AUV Collaborative System for Underwater Tasks in Extreme Sea Conditions." *IEEE Transactions on Mobile Computing*, **Accepted**, 2026. *(Peer-reviewed)*

[J3] Xiaobing Chen, Boyang Zhang, Xiangwei Zhou, Mingxuan Sun, **Shuai Zhang**, Songyang Zhang, Geoffrey Ye Li. "Efficient Training of Large-Scale AI Models Through Federated Mixture-of-Experts: A System-Level Approach." *IEEE Communications Magazine*, **Accepted**, 2025. *(Peer-reviewed)*

### B.2 Published

[J4] Hongkang Li, **Shuai Zhang**, Yihua Zhang, Meng Wang, Sijia Liu, Pin-Yu Chen. "How Does Promoting the Minority Fraction Affect Generalization? A Theoretical Study of One-Hidden-Layer Neural Networks on Group Imbalance." *IEEE Journal of Selected Topics in Signal Processing*, vol. 18, no. 2, pp. 216–231, 2024. *(Peer-reviewed)*

[J5] Jingzehua Xu, Guanwen Xie, Zekai Zhang, Xiangwang Hou, **Shuai Zhang**, Yong Ren, Dusit Niyato. "UPEGSim: An RL-Enabled Simulator for Unmanned Underwater Vehicles Dedicated to the Underwater Pursuit–Evasion Game." *IEEE Internet of Things Journal*, vol. 12, no. 3, pp. 2334–2346, 2025. *(Peer-reviewed)*

[J6] Guanwen Xie, Jingzehua Xu, Ziqi Zhang, Xiangwang Hou, Dongfang Ma, **Shuai Zhang**, Yong Ren, Dusit Niyato. "Is Fisher All You Need in the Multi-AUV Underwater Target Tracking Task?" *IEEE Transactions on Mobile Computing*, 2025. *(Peer-reviewed)*

[J7] **Shuai Zhang**, Meng Wang. "Correction of Corrupted Columns through Fast Robust Hankel Matrix Completion." *IEEE Transactions on Signal Processing*, vol. 67, no. 10, pp. 2580–2594, 2019. *(Peer-reviewed)*

[J8] **Shuai Zhang**, Meng Wang, Jinjun Xiong, Sijia Liu, Pin-Yu Chen. "Improved Linear Convergence of Training CNNs with Generalizability Guarantees: A One-Hidden-Layer Case." *IEEE Transactions on Neural Networks and Learning Systems*, vol. 32, no. 6, pp. 2622–2635, 2020. *(Peer-reviewed)*

[J9] **Shuai Zhang**, Yingshuai Hao, Meng Wang, Joe H. Chow. "Multichannel Hankel Matrix Completion through Nonconvex Optimization." *IEEE Journal of Selected Topics in Signal Processing*, vol. 12, no. 4, pp. 617–632, 2018. *(Peer-reviewed)*

## C. Refereed Conference (Main Track) Papers

### C.1 Top ML/AI Conferences (ICML, ICLR, NeurIPS)

[C1] Hongkang Li, Yihua Zhang, **Shuai Zhang**, Meng Wang, Sijia Liu, Pin-Yu Chen. "When Is Task Vector Provably Effective for Model Editing? A Generalization Analysis of Nonlinear Transformers." *International Conference on Learning Representations (ICLR)*, **Oral**, 2025. *(Peer-reviewed, Oral acceptance: Top 1.8%)*

[C2] Nowaz Chowdhury, **Shuai Zhang**, Meng Wang, Sijia Liu, Pin-Yu Chen. "Patch-Level Routing in Mixture-of-Experts Is Provably Sample-Efficient for Convolutional Neural Networks." *International Conference on Machine Learning (ICML)*, **Oral**, 2023. *(Peer-reviewed, Oral acceptance: Top 2.37%)*

[C3] **Haixu Liao***, Yating Zhou, Songyang Zhang, Meng Wang, **Shuai Zhang**. "Theoretical Analysis of Contrastive Learning under Imbalanced Data: From Training Dynamics to a Pruning Solution." *International Conference on Learning Representations (ICLR)*, 2026. *(Peer-reviewed, ∼28% acceptance)*

[C4] **Mugunthan Shandirasegaran***, Hongkang Li, Songyang Zhang, Meng Wang, **Shuai Zhang**. "A Theoretical Analysis of Mamba's Training Dynamics: Filtering Relevant Features for Generalization in State Space Models." *International Conference on Learning Representations (ICLR)*, 2026. *(Peer-reviewed, ∼28% acceptance)*

[C5] Jiawei Sun, **Shuai Zhang**, Hongkang Li, Meng Wang. "Contrastive Learning with Data Misalignment: Feature Purity, Training Dynamics, and Theoretical Generalization Guarantees." *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. *(Peer-reviewed, 24.52% acceptance)*

[C6] Zixi Wang, Yushe Cao, Yubo Huang, Jinzhu Wei, Jingzehua Xu, **Shuai Zhang**, Xin Lai. "Self-Training with Dynamic Weighting for Robust Gradual Domain Adaptation." *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. *(Peer-reviewed, 24.52% acceptance)*

[C7] **Shuai Zhang**, Heshan Devaka Fernando, Miao Liu, Keerthiram Murugesan, Songtao Lu, Pin-Yu Chen, Tianyi Chen, Meng Wang. "SF-DQN: Provable Knowledge Transfer Using Successor Features for Deep Reinforcement Learning." *International Conference on Machine Learning (ICML)*, 2024. *(Peer-reviewed, 27.5% acceptance)*

[C8] Shusen Jing, Anlan Yu, **Shuai Zhang**, Songyang Zhang. "FedSC: Provable Federated Self-Supervised Learning with Spectral Contrastive Objective over Non-IID Data." *International Conference on Machine Learning (ICML)*, 2024. *(Peer-reviewed, 27.5% acceptance)*

[C9] **Shuai Zhang**, Hongkang Li, Meng Wang, Miao Liu, Pin-Yu Chen, Songtao Lu, Sijia Liu, Keerthiram Murugesan, Subhajit Chaudhury "On the Convergence and Sample Complexity Analysis of Deep Q-Networks with $\epsilon$-Greedy Exploration." *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. *(Peer-reviewed, 30% acceptance)*

[C10] **Shuai Zhang**, Meng Wang, Pin-Yu Chen, Sijia Liu, Songtao Lu, Miao Liu. "Joint Edge–Model Sparse Learning Is Provably Efficient for Graph Neural Networks." *International Conference on Learning Representations (ICLR)*, 2023. *(Peer-reviewed, 31.79% acceptance)*

[C11] **Shuai Zhang**, Meng Wang, Sijia Liu, Pin-Yu Chen, Jinjun Xiong. "How Unlabeled Data Improve Generalization in Self-Training? A One-Hidden-Layer Theoretical Analysis." *International Conference on Learning Representations (ICLR)*, 2022. *(Peer-reviewed, 32% acceptance)*

[C12] **Shuai Zhang**, Meng Wang, Sijia Liu, Pin-Yu Chen, Jinjun Xiong. "Why Lottery Ticket Wins? A Theoretical Perspective of Sample Complexity on Sparse Neural Networks." *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, pp. 2707–2720, 2021. *(Peer-reviewed, 26.1% acceptance)*

[C13] **Shuai Zhang**, Meng Wang, Sijia Liu, Pin-Yu Chen, Jinjun Xiong. "Fast Learning of Graph Neural Networks with Guaranteed Generalizability: One-Hidden-Layer Case." *International Conference on Machine Learning (ICML)*, pp. 11268–11277, 2020. *(Peer-reviewed, 21.8% acceptance)*

### C.2 Selective AI and Applied AI Conferences (e.g., AISTATS, IROS, ACMMM)

[C14] Yihua Zhang, Hongkang Li, Yuguang Yao, Aochuan Chen, **Shuai Zhang**, Pin-Yu Chen, Meng Wang, Sijia Liu. "Visual Prompting Reimagined: The Power of Activation Prompts." *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2026. *(Peer-reviewed, ∼25% acceptance)*

[C15] Zixi Wang, Yubo Huang, Jingzehua Xu, Jinzhu Wei, **Shuai Zhang**, Xin Lai. "Multi-Modal Gradual Domain Osmosis: Stepwise Dynamic Learning with Batch Matching for Gradual Domain Adaptation." *33rd ACM International Conference on Multimedia (ACMMM)*, 2025. *(Peer-reviewed, 23.5% accep-*

*tance)*

[C16] Yimian Ding, Jingzehua Xu, Guanwen Xie, **Shuai Zhang**, Yi Li. "Make Your AUV Adaptive: An Environment-Aware Reinforcement Learning Framework for Underwater Tasks." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. *(Peer-reviewed, ∼46% acceptance)*

[C17] Guanwen Xie, Jingzehua Xu, Yimian Ding, Zhi Zhang, **Shuai Zhang**, Yi Li. "Never Too Prim to Swim: An LLM-Enhanced RL-Based Adaptive Sea-Surface Controller for AUVs under Extreme Sea Conditions." *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025. *(Peer-reviewed, ∼46% acceptance)*

### C.3 Selective Signal Processing and Communications Conferences (e.g., ICASSP, ISIT)

[C18] Yubo Huang, Xin Lai, Muyang Ye, Anran Zhu, Zixi Wang, Jingzehua Xu, **Shuai Zhang**, Zhiyuan Zhou, Weijie Niu. "LHQ-SVC: Lightweight and High-Quality Singing Voice Conversion Modeling." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025. *(Peer-reviewed, 45.27% acceptance)*

[C19] Jingzehua Xu, Guanwen Xie, Xinqi Wang, Yimian Ding, **Shuai Zhang**. "USV–AUV Collaboration Framework for Underwater Tasks under Extreme Sea Conditions." *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025. *(Peer-reviewed, 45.27% acceptance)*

[C20] **Shuai Zhang**, Meng Wang, Sijia Liu, Pin-Yu Chen, Jinjun Xiong. "Guaranteed Convergence of Training Convolutional Neural Networks via Accelerated Gradient Descent." *Conference on Information Sciences and Systems (CISS)*, pp. 1–6, 2020. *(Peer-reviewed, 15% acceptance)*

[C21] Hongkang Li, **Shuai Zhang**, Meng Wang. "Learning and Generalization of One-Hidden-Layer Neural Networks, Going Beyond Standard Gaussian Data." *Conference on Information Sciences and Systems (CISS)*, pp. 37–42, 2022. *(Peer-reviewed, 20.5% acceptance)*

[C22] Meng Wang, Joe H. Chow, Yingshuai Hao, **Shuai Zhang**, Wenting Li, Ren Wang, Pengzhi Gao, Christopher Lackner, Evangelos Farantatos, Mahendra Patel. "A Low-Rank Framework of PMU Data Recovery and Event Identification." *IEEE International Conference on Smart Grid Synchronized Measurements and Analytics (SGSMA)*, pp. 1–9, 2019. *(Peer-reviewed)*

[C23] **Shuai Zhang**, Meng Wang. "Correction of Simultaneous Bad Measurements by Exploiting the Low-Rank Hankel Structure." *IEEE International Symposium on Information Theory (ISIT)*, pp. 646–650, 2018. *(Peer-reviewed, ∼50% acceptance)*

## D. Peer-Reviewed Workshop Proceedings

[W1] **Mugunthan Shandirasegaran***, Yating Zhou, Songyang Zhang, **Shuai Zhang**. "Theoretical Analysis of the Selection Mechanism in Mamba: Training Dynamics and Generalization." *NeurIPS 2025 Workshop*, 2025. *(Peer-reviewed; acceptance rate not publicly available)*

[W2] **Haixu Liao***, Yating Zhou, Songyang Zhang, **Shuai Zhang**. "On the Training Dynamics of Contrastive Learning with Imbalanced Feature Distributions: A Theoretical Study of Feature Learning." *NeurIPS 2025 Workshop*, 2025. *(Peer-reviewed)*

[W3] Qiuchen Song, Shusen Jing, **Shuai Zhang**, Songyang Zhang, Chuan Huang. "Mixture-of-Experts for Distributed Edge Computing with Channel-Aware Gating Function." *IEEE International Conference on Communications Workshops (ICC Workshops)*, 2025. *(Peer-reviewed)*

[W4] Hongkang Li, Meng Wang, **Shuai Zhang**, Sijia Liu, Pin-Yu Chen. "Learning on Transformers Is Provable Low-Rank and Sparse: A One-Layer Analysis." *IEEE Sensor Array and Multichannel Signal*

*Processing Workshop (SAM)*, pp. 1–5, 2024. *(Peer-reviewed)*

[W5] **Shuai Zhang**, Meng Wang, Joe H. Chow. "Multi-Channel Missing Data Recovery by Exploiting the Low-Rank Hankel Structures." *IEEE Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pp. 1–5, 2017. *(Peer-reviewed)*

[W6] Jingzehua Xu, Guanwen Xie, Yimian Ding, Yongming Zeng, Haoyu Wang, **Shuai Zhang**. "UACOF: A USV–AUV Collaboration Framework for Underwater Tasks under Extreme Sea Conditions (Student Abstract)." *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 28, pp. 29538–29540, 2025. *(Peer-reviewed)*

[W7] Yimian Ding, Jingzehua Xu, Yiyuan Yang, Guanwen Xie, Xinqi Wang, **Shuai Zhang**. "AoI-MDP: An AoI-Optimized Markov Decision Process Dedicated to Underwater Tasks (Student Abstract)." *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 28, pp. 29348–29350, 2025. *(Peer-reviewed)*

[W8] Guanwen Xie, Jingzehua Xu, Yiyuan Yang, Yimian Ding, **Shuai Zhang**. "ErfsI: An Efficient Reward Function Searcher via Large Language Models for Custom-Environment Multi-Objective Reinforcement Learning (Student Abstract)." *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 28, pp. 29535–29537, 2025. *(Peer-reviewed)*

## Grants

### Awarded Grants

- **NSF**, Principal Investigator. *Collaborative Research: CSR: Small: Interpretable and Efficient Signal Processing and Deep Learning over Graphs for Spatio-Temporal Data Analysis in AIoT Systems.* 03/2025 – 02/2028. $200,000 (NJIT share) of $400,000 total.

### Grant Proposals Pending

- **NSF**, Principal Investigator. *Collaborative Research: VINES Track 1: Hybrid-Field Communications for Low-Altitude Economy with AI-Native Digital Twinning.* Submitted 08/2025. Proposed duration: 3 years. $299,875 (NJIT share) of $1,500,000 total.
- **NSF**, Principal Investigator. *Collaborative Research: MFAI: Mathematical and Algorithmic Foundations of Large Language Models: Training, Generalization, Modularization.* Submitted 10/2025. Proposed duration: 3 years. $374,444 (NJIT share) of $1,500,000 total.
- **NSF**, Principal Investigator. *Collaborative Research: VINES Track 1: AI-Empowered Semantic Communications with Energy Efficiency and Privacy Preserving for IoT-Enabled Emergency Response.* Submitted 08/2025. Proposed duration: 3 years. $369,929 (NJIT share) of $1,500,000 total.

## Teaching

- **Machine Learning (DS 675):** Fall 2023, Spring 2025, Fall 2025
- **Advanced Machine Learning (CS 732):** Spring 2024, Fall 2024
- **Foundations of Machine Learning (DS 400):** Spring 2026

## Advising and Mentoring

### Doctoral Students (Primary Advisor)

- Haixu Liao (Data Science, NJIT), 2024–Present. Expected 2029.
- Mugunthan Shandirasegaran (Data Science, NJIT), 2024–Present. Expected 2029.

**Master's Students (Primary Advisor)**

- Midde, Gnana Abhishek (Data Science, NJIT), Fall 2025–Present. Expected Spring 2027.
- Thomas, Anson (Data Science, NJIT), Spring 2024–Fall 2024.
- Notla, Preetham (Data Science, NJIT), Spring 2024–Fall 2024.

**Undergraduate Student (Primary Advisor)**

- Marcus Hilario (Computer Science, NJIT), Sep. 2025–Present. Project: *Optimizer Selection in Machine Learning Models.*

**Doctoral Dissertation Committee Service**

- Vidal Agustin (Data Science, NJIT), April 2026.
- Zhibo Ye (Business, NJIT), March 2025.
- Shaoze Fan (Computer Science, NJIT), April 2025.

**Other Supervision / Research Mentoring**

- Neranjan Senarath (EE, RPI), Fall 2026-Spring 2026.
- Jiawei Sun (EE, RPI), Fall 2024-Spring 2026.
- Hongkang Li (EE, RPI), Fall 2023 - Fall 2024.
- Jingzehua Xu (Visiting Student at MIT), Fall 2023-Fall 2024.
- Weihang Zhang (ECE, Duke University), Fall 2025.

## Invited and Conference Presentations

| | |
|---|---|
| **Invited Talk** | [**ECE Seminar@Illinois Tech**] Towards Theoretical Understandings of Foundation models: Feature Learning, Optimization Dynamics, and Generalization. |
| **Invited Talk** | [**SINE Lab@Cornell Tech**] Towards Theoretical Understandings of Foundation models: Feature Learning, Optimization Dynamics, and Generalization. |
| **Invited Talk** | [**INFORMS Annual Mtg 2024**] Provable Efficient Graph Neural Network Learning via Joint Edge-Model Sparsification. |
| **Invited Talk** | [**NSF CSR PI 2025**] Interpretable and Efficient Signal Processing and Deep Learning over Graphs for Spatio-Temporal Data Analysis in AIoT Systems |
| **Oral** | [**ICLR 2025**] When Is Task Vector Provably Effective for Model Editing? A Generalization Analysis of Nonlinear Transformers. |
| **Oral** | [**ICML 2023**] Patch-Level Routing in Mixture-of-Experts Is Provably Sample-Efficient for Convolutional Neural Networks. |
| **Poster** | [**ICLR 2026**] Theoretical Analysis of Contrastive Learning under Imbalanced Data: From Training Dynamics to a Pruning Solution |
| **Poster** | [**ICLR 2026**] A Theoretical Analysis of Mamba's Training Dynamics: Filtering Relevant Features for Generalization in State Space Models |
| **Poster** | [**NeurIPS 2025**] Contrastive Learning with Data Misalignment: Feature Purity, Training Dynamics and Theoretical Generalization Guarantees |

| | |
|---|---|
| **Poster** | [**NeurIPS 2025**] Self-Training with Dynamic Weighting for Robust Gradual Domain Adaptation. |
| **Poster** | [**ACMMM 2025**] Multi-Modal Gradual Domain Osmosis: Stepwise Dynamic Learning with Batch Matching for Gradual Domain Adaptation |
| **Poster** | [**IROS 2025**] Never Too Prim to Swim: An LLM-Enhanced RL-Based Adaptive S-Surface Controller for AUVs under Extreme Sea Conditions. |
| **Poster** | [**IROS 2025**] Make Your AUV Adaptive: An Environment-Aware Reinforcement Learning Framework for Underwater Tasks. |
| **Poster** | [**ICML 2024**] SF-DQN: Provable Knowledge Transfer Using Successor Feature for Deep Reinforcement Learning. |
| **Poster** | [**NeurIPS 2023**] On the Convergence and Sample Complexity Analysis of Deep Q-Networks with $\varepsilon$-Greedy Exploration. |
| **Poster** | [**ICML 2024**] FedSC: Provable Federated Self-supervised Learning with Spectral Contrastive Objective over Non-iid Data. |
| **Poster** | [**ICLR 2023**] Joint Edge-Model Sparse Learning Is Provably Efficient for Graph Neural Networks. |
| **Poster** | [**ICLR 2022**] How Unlabeled Data Improve Generalization in Self-Training? A One-Hidden-Layer Theoretical Analysis. |
| **Poster** | [**NeurIPS 2021**] Why Lottery Ticket Wins? A Theoretical Perspective of Sample Complexity on Sparse Neural Networks. |
| **Poster** | [**ICML 2020**] Fast Learning of Graph Neural Networks with Guaranteed Generalizability: One-Hidden-Layer Case. |

## Service Activities

### Editorial Activity

- Guest Editor, special issue "Application of Artificial Intelligence to Image Processing: Advantages and Prognosis," *Electronics (MDPI)*.
- Guest Editor, special issue "Signal Processing for Representation Learning: Theoretical Foundations and Emerging Applications," *Frontiers in Signal Processing*.

### Professional Reviewing and Program Committee Service

- Panelist, National Science Foundation (NSF), Computer and Information Science and Engineering (CISE), Apr. 2026.
- Area Chair, Conference on Neural Information Processing Systems (NeurIPS), 2025, 2026.
- Area Chair, International Conference on Learning Representations (ICLR), 2026.
- Conference reviewer, Conference on Neural Information Processing Systems (NeurIPS), 2023–2025
- Conference reviewer, International Conference on Learning Representations (ICLR), 2023–2025
- Conference reviewer, International Conference on Machine Learning (ICML), 2023–2026
- Conference reviewer, International Conference on Artificial Intelligence and Statistics (AISTATS), 2025
- Journal reviewer, Transactions on Machine Learning Research (TMLR)
- Conference reviewer, IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, 2026

- Journal reviewer, IEEE Transactions on Signal Processing
- Journal reviewer, IEEE Transactions on Information Theory
- Conference reviewer, the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2021-2025
- Journal reviewer, The VLDB Journal